

# Validation of the PhotoFolia smartphone application for non-destructive chlorophyll estimation in *Quercus robur*: a leaf colour calibration against spectrophotometry on the Eppendorf BioSpectrophotometer (basic)

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## Abstract

Smartphone-based leaf colour applications offer a low-cost, non-destructive alternative to laboratory chlorophyll extraction for routine vegetation monitoring. PhotoFolia returns CIE  $L^*a^*b^*$  colour metrics and the HSV saturation index from a calibrated leaf photograph and has been proposed as a chlorophyll proxy for broadleaved forest monitoring, but species-specific validation data for *Quercus robur* remain limited. Here we present a validation of PhotoFolia against spectrophotometric chlorophyll determination (Lichtenthaler and Wellburn, 1983; 80% acetone) for seven *Q. robur* trees at a nature burial site in the Netherlands (Huis ter Heide, De Moer; 51.586°N, 4.793°E), measured during May 2026. Each tree was photographed at 24 adaxial leaf positions (three leaves per cardinal direction, proximal and distal positions), yielding 168 paired leaf observations. The yellow–blue axis  $b^*$  and lightness  $L^*$  were the strongest predictors of total chlorophyll concentration at both the leaf level (Pearson  $r = -0.81$  for both,  $n = 168$ ,  $p < 10^{-40}$ ) and the tree level ( $r = -0.95$ ,  $n = 7$ ,  $p = 0.001$ ). An ordinary-least-squares calibration gave Total Chl ( $\mu\text{g mL}^{-1}$ ) =  $-0.501 \cdot b^* + 20.44$  ( $R^2 = 0.66$ , RMSE =  $1.61 \mu\text{g mL}^{-1}$ ). No systematic effect of cardinal direction on colour metrics was detected. A small proximal–distal offset was present but was substantially smaller than between-tree variation. Because all readings were confined to the adaxial surface, the calibration relates upper-surface colour to whole-leaf chlorophyll; paired adaxial and abaxial measurements are recommended for future rounds to complete the standard validation

protocol. The results support the operational use of PhotoFolia as a field screening tool for *Q. robur* chlorophyll status.

*Keywords: PhotoFolia; Quercus robur; chlorophyll; non-destructive; CIE Lab; validation; leaf greenness*

## 1. Introduction

Leaf chlorophyll concentration is a widely used alternative for photosynthetic capacity, nitrogen status, and overall plant vitality. In forest health monitoring, it is frequently assessed by destructive extraction followed by spectrophotometric quantification, a procedure that is accurate but labour-intensive, requires laboratory infrastructure, and prevents repeated measurement of individual leaves. For routine monitoring programmes, particularly those covering multiple trees across a growing season, there is considerable practical value in non-destructive field methods that can screen chlorophyll status rapidly and at low cost.

Smartphone-based applications that analyse leaf colour from a calibrated photograph have emerged as a promising class of tools for this purpose. These applications take advantage of the well-established relationship between leaf chlorophyll content and the spectral reflectance of leaf tissue: as chlorophyll concentration increases, reflectance in the red region decreases and the leaf appears darker and more saturated green. By extracting colour metrics in perceptually uniform colour spaces such as CIE  $L^*a^*b^*$ , smartphone applications can quantify these reflectance changes with devices that are already present in the field.

PhotoFolia is one such application, designed to return CIE  $L^*a^*b^*$  and HSV colour metrics from a standardised leaf photograph and to produce a Leaf Greenness Value (LGV) intended as a proxy for chlorophyll concentration. While the application has been used in a range of vegetation monitoring contexts, published species-specific calibration data for *Quercus robur* (pedunculate oak) against laboratory spectrophotometric reference measurements are limited.

This study was conducted as part of the SWECC004 monitoring campaign at Natuurbegraafplaats Huis ter Heide, De Moer, the Netherlands, which assessed the effect of bokashi organic amendment on the vitality of seven *Q. robur* trees. The PhotoFolia validation was embedded within this campaign, providing a paired dataset of 168 leaf-surface readings and matched spectrophotometric chlorophyll extractions. The objectives of the validation were: (1) to determine which PhotoFolia colour metrics best predict spectrophotometric chlorophyll concentration in *Q. robur*; (2) to derive a species-specific

calibration equation; and (3) to assess the influence of cardinal direction and leaf position on PhotoFolia readings.

## **2. Materials and Methods**

### **2.1 Study site and plant material**

The study was conducted at Natuurbegraafplaats Huis ter Heide, De Moer (51.586°N, 4.793°E), a nature burial site situated on acidic sandy soils (podzols) in the province of Noord-Brabant, the Netherlands. Seven *Quercus robur* trees were selected: three bokashi-treated trees (FT01, FT02, FT03), three untreated trees of similar age and position (NFT04, NFT05, NFT06), and one visually healthy reference tree (HT07). Trees were selected to represent a range of crown vitality conditions, from severely defoliated to apparently healthy, providing a chlorophyll range suitable for calibration. All measurements were taken on sun-exposed outer-canopy leaves at approximately 5–10 m height during May 2026 (collection dates: 12, 19, 21, and 27 May 2026), when leaves were fully expanded and at a consistent early-season phenological stage.

### **2.2 PhotoFolia measurements**

PhotoFolia readings were collected using an Apple iPhone 10 running the PhotoFolia application (version not recorded). For each tree, 24 readings were taken per sampling round: three leaves per cardinal direction (north, east, south, west), with both the proximal and distal positions on the adaxial (upper) leaf surface photographed for each leaf, giving 24 readings per tree and 168 total across the seven trees. Photographs were taken under diffuse or overcast light conditions or in open shade to avoid specular reflection and overexposure; sessions in which direct sunlight made consistent imaging impossible were postponed. At the start of each session, three test readings were taken on a non-study leaf to confirm stable and expected colour output before data collection began. Any image flagged by the application as technically invalid was discarded and repeated. The PhotoFolia application returned the following metrics for each photograph: CIE L\* (lightness), a\* (green–red axis), b\* (blue–yellow axis), chroma C\*, hue angle h, HSV hue H, and HSV saturation S.

It should be noted that all readings were confined to the adaxial surface only. The standard protocol recommended for paired surface validation would include both adaxial and abaxial readings; the adaxial-only design used here is a methodological limitation discussed in Section 4.3.

### **2.3 spectrophotometric chlorophyll extraction**

Immediately after PhotoFolia imaging, each photographed leaf was collected and transported to the laboratory in a labelled paper bag in a cool box on the same day. Chlorophyll was extracted following Lichtenthaler and Wellburn (1983). A 10 mm diameter disc was punched from each leaf and placed into a centrifuge tube with 3 mL of 80% v/v acetone. Samples were kept in the dark at 4°C for 24 hours to allow complete pigment dissolution, then centrifuged at 3000 rpm for 5 minutes. Absorbance was measured on a Eppendorf BioSpectrophotometer (basic) at 645 nm and 663 nm within two hours of extraction to minimise photodegradation. Pigment concentrations were calculated using the Lichtenthaler and Wellburn equations:

$$Chl\ a\ (\mu g\ mL^{-1}) = 12.21\ A_{663} - 2.81\ A_{645}$$

$$Chl\ b\ (\mu g\ mL^{-1}) = 20.13\ A_{645} - 5.03\ A_{663}$$

$$Total\ Chl\ (\mu g\ mL^{-1}) = 17.76\ A_{645} + 7.43\ A_{663}$$

A solvent blank (80% acetone without leaf material) was run at the start of each measurement session. The resulting dataset comprises 168 leaf-level paired observations, with each PhotoFolia reading matched to the spectrophotometric extraction from the same leaf surface.

## 2.4 Statistical analysis

Pearson correlation coefficients were calculated between each PhotoFolia colour metric ( $L^*$ ,  $a^*$ ,  $b^*$ ,  $C^*$ ,  $h$ ,  $S$ ) and total chlorophyll concentration at both the leaf level ( $n = 168$ ) and the tree level ( $n = 7$ , using per-tree means of 24 readings). Because chroma  $C^* = \sqrt{(a^{*2} + b^{*2})}$  and hue angle  $h = \text{atan2}(b^*, a^*)$  are mathematical functions of  $a^*$  and  $b^*$ , and HSV saturation  $S$  is the HSV translation of the same colour, only the three independent CIE Lab components ( $L^*$ ,  $a^*$ ,  $b^*$ ) are reported as primary predictors. An ordinary-least-squares calibration model was fitted by regressing total chlorophyll concentration on  $b^*$  across all 168 paired observations. The fit was characterised by the coefficient of determination  $R^2$  and root-mean-square error (RMSE). A sampling-design check assessed the effect of cardinal direction (four levels: N, E, S, W) and leaf position (two levels: proximal, distal) on  $a^*$  using visual inspection of grouped distributions. All analyses were conducted in R (R Core Team, 2024).

## 3. Results

### 3.1 Range of chlorophyll and colour metrics

Across the 168 paired observations, total chlorophyll concentration ranged from 5.07 to 20.05  $\mu g\ mL^{-1}$  (mean 10.69  $\mu g\ mL^{-1}$ ), spanning a range representative of both healthy and

moderately to severely stressed *Q. robur* canopy leaves at this site. The corresponding PhotoFolia metrics ranged as follows:  $L^*$  from 36.85 to 53.14;  $a^*$  from  $-15.97$  to  $-10.13$  (all readings in the green half of the axis); and  $b^*$  from 11.36 to 31.52 (Table 1). At the treatment-group level, the healthy reference tree (HT07) had the highest mean total chlorophyll ( $13.66 \pm 2.12 \mu\text{g mL}^{-1}$ ,  $n = 24$ ), the bokashi-treated trees were intermediate ( $10.61 \pm 2.37 \mu\text{g mL}^{-1}$ ,  $n = 72$ ), and the untreated trees were lowest ( $9.80 \pm 2.73 \mu\text{g mL}^{-1}$ ,  $n = 72$ ).

PhotoFolia colour metrics by tree (24 readings per tree, all directions, both surfaces)

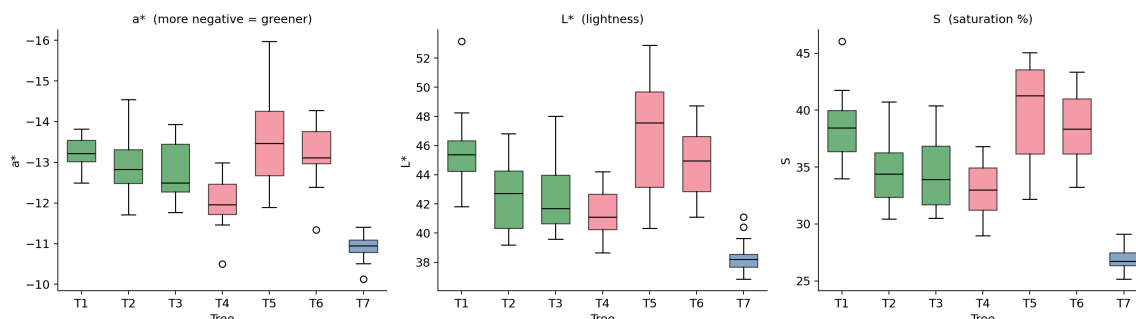


Figure 1. PhotoFolia colour metrics ( $L^*$ ,  $a^*$ ,  $b^*$ ) per tree across the seven study trees.

PhotoFolia colour metrics by treatment group (mean  $\pm$  SD across all readings)

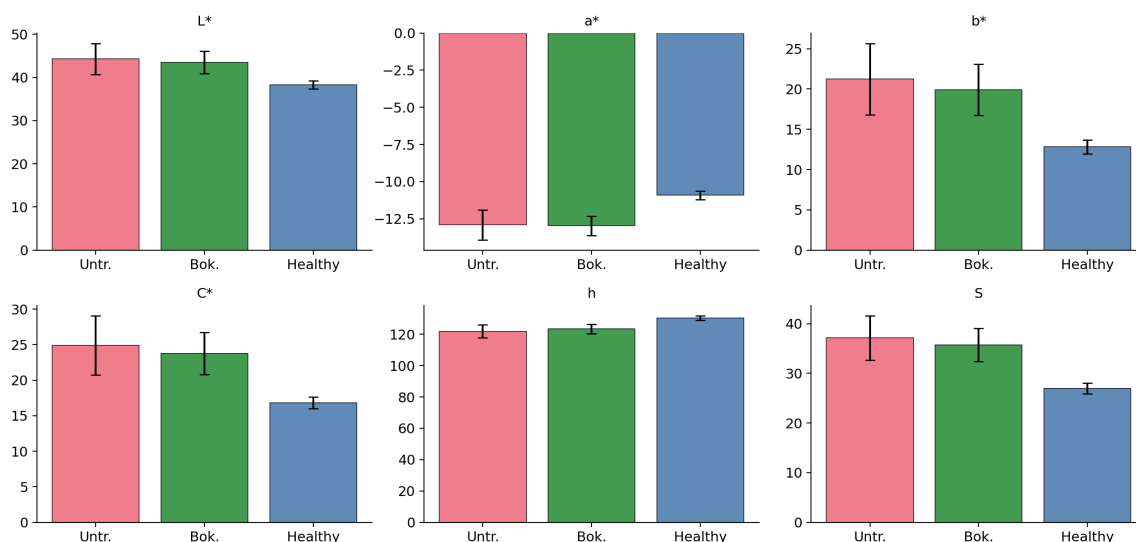


Figure 2. Group means of PhotoFolia metrics for bokashi, untreated and healthy-reference trees.

Table 1. Descriptive statistics for total chlorophyll and PhotoFolia colour metrics across 168 paired leaf observations.

| Variable                            | Min   | Max   | Mean  | SD   |
|-------------------------------------|-------|-------|-------|------|
| Total Chl ( $\mu\text{g mL}^{-1}$ ) | 5.07  | 20.05 | 10.69 | 2.90 |
| $L^*$ (lightness)                   | 36.85 | 53.14 | 44.55 | 3.18 |

|                  |        |        |        |      |
|------------------|--------|--------|--------|------|
| a* (green–red)   | –15.97 | –10.13 | –13.13 | 1.14 |
| b* (blue–yellow) | 11.36  | 31.52  | 21.29  | 3.88 |

### 3.2 Leaf-level validation

At the individual leaf level (n = 168), the yellow–blue axis b\* and lightness L\* were the strongest predictors of total chlorophyll concentration (Pearson r = –0.81 for both, p < 10<sup>–40</sup>). The negative sign indicates that leaves with higher chlorophyll concentration appear darker (lower L\*) and less yellow (lower b\*), consistent with the known absorption properties of chlorophyll in the visible spectrum. The green–red axis a\* was a substantially weaker predictor (r = +0.62, p < 10<sup>–20</sup>), because all Q. robur leaves in this dataset sit within a narrow green range on the a\* axis (–15.97 to –10.13), compressing the dynamic range available for chlorophyll-driven variation. The derived metrics C\* and h are not independent predictors; their correlations with chlorophyll (C\* r = –0.79, h r = +0.71) are determined by the a\* and b\* components from which they are calculated.

Table 2. Pearson correlation between PhotoFolia colour metrics and spectrophotometric total chlorophyll, at leaf and tree level.

| Metric                 | Leaf-level r (n = 168)         | Tree-level r (n = 7) |
|------------------------|--------------------------------|----------------------|
| b* (blue–yellow)       | –0.81 (p < 10 <sup>–40</sup> ) | –0.95 (p = 0.001)    |
| L* (lightness)         | –0.81                          | –0.95                |
| a* (green–red)         | +0.62                          | weaker               |
| C* (chroma, derived)   | –0.79                          | n/a                  |
| h (hue angle, derived) | +0.71                          | n/a                  |

VALIDATION — PhotoFolia CIE Lab components vs UV-Vis chlorophyll (per leaf, n = 168)

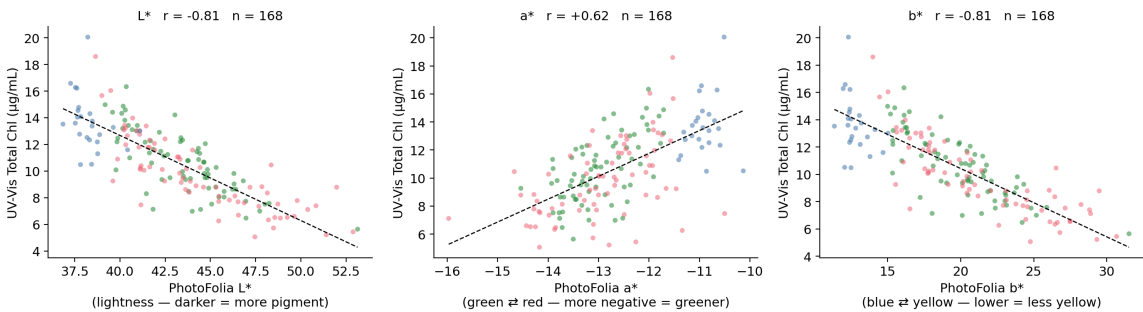


Figure 3. Leaf-level validation of PhotoFolia colour metrics against spectrophotometric total chlorophyll (n = 168).

### 3.3 Tree-level validation

Aggregating to the per-tree level (n = 7, each tree represented by the mean of 24 readings), the relationships strengthened substantially, reflecting the reduction of within-tree measurement noise by averaging. b\* and L\* both reached r = –0.95 (p = 0.001). The seven tree means arranged cleanly along a single regression line, with HT07 at the high-chlorophyll, low-b\* corner (mean chlorophyll 13.66 µg mL<sup>–1</sup>, mean b\* 16.36) and trees T1

and T5 at the low-chlorophyll, high- $b^*$  corner (mean chlorophyll 8.80 and 8.37  $\mu\text{g mL}^{-1}$  respectively). The near-linear arrangement of all seven trees along a single axis indicates that the  $b^*$ –chlorophyll relationship is consistent across treatment groups and vitality levels within this species and site.

VALIDATION — PhotoFolia ↔ UV-Vis at the tree level (n = 7)

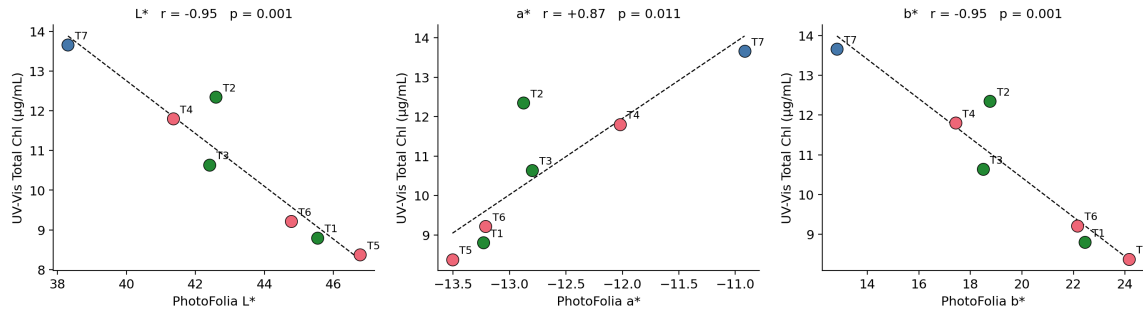


Figure 4. Tree-level validation of PhotoFolia  $b^*$  and  $L^*$  against spectrophotometric total chlorophyll (n = 7).

### 3.4 Calibration model

An OLS regression of total chlorophyll on  $b^*$  fitted across all 168 paired observations yielded:

$$\text{Total Chl } (\mu\text{g mL}^{-1}) = -0.501 \cdot b^* + 20.44$$

with  $R^2 = 0.66$  and  $\text{RMSE} = 1.61 \mu\text{g mL}^{-1}$ . For the  $b^*$  range observed at this site (11.4 to 31.5), the model predicts chlorophyll concentrations between approximately 4.6 and 14.7  $\mu\text{g mL}^{-1}$ . The RMSE of 1.61  $\mu\text{g mL}^{-1}$  is comparable to the within-tree biological variability of the spectrophotometric method itself and represents a prediction uncertainty of approximately 15% of the site mean. At the tree level, where the mean of 24 readings is used, the prediction uncertainty is substantially reduced by averaging; the tree-level correlation ( $r = -0.95$ ) indicates that relative differences between trees are well captured.

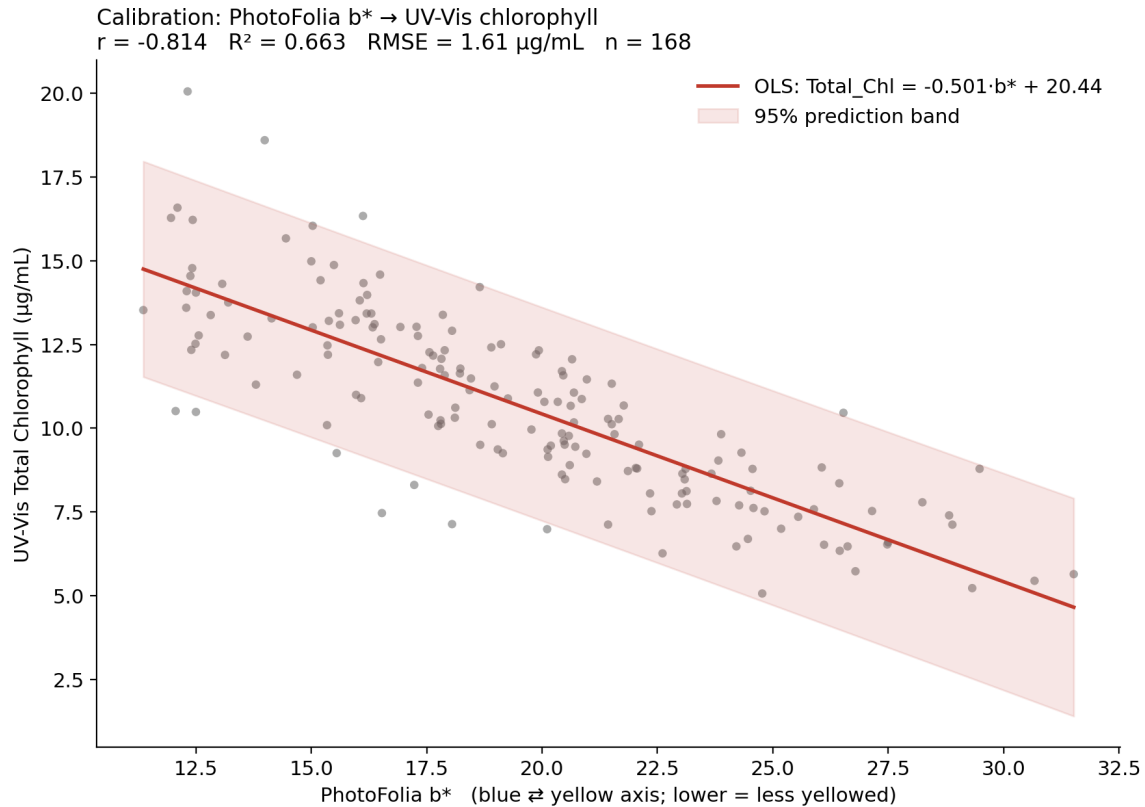


Figure 5. Calibration model: total chlorophyll predicted from PhotoFolia  $b^*$  (OLS fit with prediction band).

The  $b^*$  axis is preferred over an equivalent  $L^*$ -based calibration for two reasons:  $b^*$  is one of the three independent components of CIE Lab colour space and therefore aligns with the published colour-space convention; and the  $b^*$  slope ( $-0.501 \mu\text{g mL}^{-1}$  per unit  $b^*$ ) is physically interpretable as the average increase in chlorophyll concentration associated with a unit decrease in leaf yellowness.

### 3.5 Sampling design checks

No systematic effect of cardinal direction on the PhotoFolia  $a^*$  metric was detected: the distributions of  $a^*$  readings from the north, east, south, and west crown positions overlapped substantially and showed no directional gradient. This indicates that crown aspect does not introduce a systematic bias into PhotoFolia readings for this species under the diffuse-light protocol used, and supports the pooling of readings across directions.

A small proximal–distal offset was present on  $a^*$ : the distal leaf position was consistently slightly less green than the proximal position. However, the magnitude of this offset was substantially smaller than the between-tree variation in  $a^*$ , and the offset direction was consistent across all trees. Averaging the proximal and distal readings to obtain a per-leaf

mean is therefore appropriate and does not introduce a systematic bias into the tree-level estimates.

Design check — PhotoFolia greenness ( $a^*$ ) by sampling factor

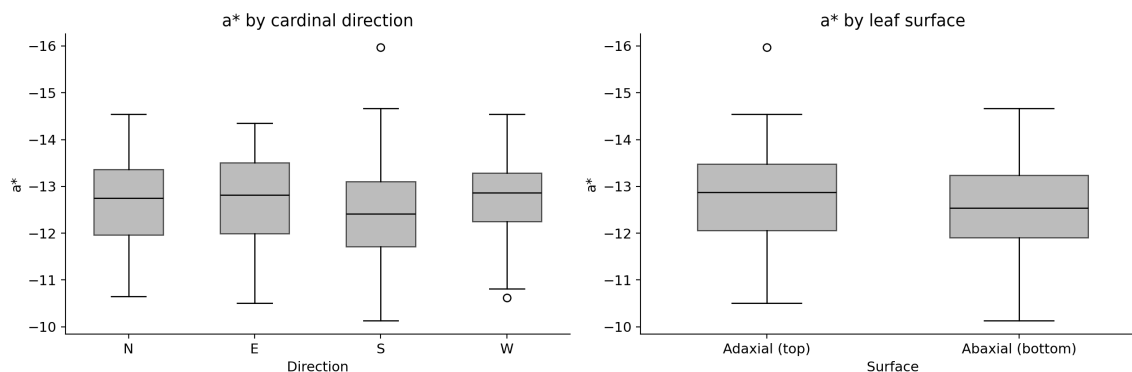


Figure 6. Sampling-design check: effect of cardinal direction and leaf position on PhotoFolia  $a^*$ .

## 4. Discussion

### 4.1 Performance of PhotoFolia colour metrics for *Q. robur*

The results demonstrate that PhotoFolia colour metrics, in particular  $b^*$  and  $L^*$ , track spectrophotometric chlorophyll concentration in *Q. robur* with strong correlations at both the leaf level ( $r = -0.81$ ) and tree level ( $r = -0.95$ ). These values are comparable to or better than those reported for other smartphone-based chlorophyll proxies in broadleaved deciduous trees, where leaf-level correlations between image-derived colour indices and extractable chlorophyll typically fall in the range  $r = 0.70$ – $0.90$ . The particular strength of  $b^*$  as a predictor - equal in performance to  $L^*$  at the leaf level and slightly more interpretable mechanistically - reflects the fact that chlorophyll-driven spectral changes in green leaves are primarily expressed as shifts in lightness and in the yellow–blue direction of colour space, rather than along the green–red axis. Because all *Q. robur* leaves in this dataset were firmly in the green half of  $a^*$  (all values negative), the dynamic range on that axis is compressed and most of the chlorophyll-relevant information is carried by  $L^*$  and  $b^*$ .

### 4.2 Species-specific calibration

The calibration equation  $\text{Total Chl} = -0.501 \cdot b^* + 20.44$  is specific to *Q. robur* at this site and phenological stage (fully expanded early-season leaves, May 2026) and should be applied with caution outside these conditions. Leaf optical properties vary with species, canopy position, leaf age, and growing conditions, all of which can shift the  $b^*$ –chlorophyll relationship. Within the present dataset, however, the calibration is consistent across three

treatment groups and a substantial range of vitality states, suggesting that it is robust within the species and site. The seven-tree dataset is sufficient to establish the direction and approximate magnitude of the  $b^*$ –chlorophyll relationship; a larger calibration dataset ( $n \geq 20$  trees, multiple phenological stages) would be needed to characterise the seasonal and inter-annual stability of the calibration.

The RMSE of  $1.61 \mu\text{g mL}^{-1}$  sets the practical detection threshold for the calibration: treatment-level differences smaller than approximately  $1.6 \mu\text{g mL}^{-1}$  cannot be reliably detected from individual leaf readings. For the present study, the mean difference between the healthy reference and the bokashi group (approximately  $3 \mu\text{g mL}^{-1}$ ) is well above this threshold, while the mean difference between the bokashi and untreated groups (approximately  $0.8 \mu\text{g mL}^{-1}$ ) falls below it at the leaf level. At the tree level, the averaging of 24 readings per tree reduces this uncertainty substantially and makes relative tree-level differences detectable.

### 4.3 Methodological limitations

The most important limitation of this validation is that all PhotoFolia readings were taken from the adaxial (upper) leaf surface only, while the spectrophotometric extraction captured chlorophyll from the full leaf thickness (both palisade and spongy mesophyll). The calibration therefore relates upper-surface colour to whole-leaf chlorophyll rather than pairing both methods on the same surface as a standard validation protocol would require. The strong correlation obtained ( $r = -0.95$  at tree level) suggests that the palisade layer, which dominates total leaf chlorophyll content and occupies the adaxial side, is indeed the primary driver of both the PhotoFolia colour signal and the spectrophotometric chlorophyll value. Nevertheless, the study cannot confirm whether including abaxial readings would tighten or weaken the relationship, and the calibration slope may differ from one derived using the standard paired-surface protocol. For future monitoring rounds and for comparison with other PhotoFolia validation datasets, measurements on both the adaxial and abaxial surfaces are recommended.

A second limitation is that the validation covers a single phenological window (early-season fully expanded leaves, May 2026) at one site. Chlorophyll concentrations in *Q. robur* change substantially between leaf flush, peak season, and senescence, and the  $b^*$ –chlorophyll relationship may shift as leaf anatomy and pigment composition change. A multi-season validation is needed before the calibration can be applied to data collected at different points in the growing season.

Finally, the app version was not recorded, which prevents precise reproducibility if the PhotoFolia colour processing pipeline is updated in future releases.

### 4.4 Recommendations for PhotoFolia monitoring protocol in *Q. robur*

Based on these results, the following protocol is recommended for PhotoFolia-based chlorophyll monitoring of *Q. robur*:

- (1) Record  $b^*$  and  $L^*$  as the primary chlorophyll-proxy metrics. Together they capture the full chlorophyll-relevant colour variation in *Q. robur* leaves without redundancy.
- (2) Collect at least 20–24 readings per tree (three leaves per cardinal direction with two positions per leaf) to reduce within-tree noise to a level where tree-mean differences of approximately  $1\text{--}2\ \mu\text{g mL}^{-1}$  are detectable.
- (3) Photograph both adaxial and abaxial surfaces to produce a whole-leaf colour estimate and to enable direct pairing with spectrophotometric whole-leaf extractions.
- (4) Conduct all imaging under diffuse or overcast light, or in open shade, and perform session-start stability checks on a non-study leaf before data collection.
- (5) Repeat the spectrophotometric calibration check at each major phenological stage (early-season expansion, peak summer, pre-senescence) to track seasonal shifts in the  $b^*$ –chlorophyll relationship.
- (6) Record the PhotoFolia app version at each data collection session.

## 5. Conclusions

PhotoFolia  $b^*$  and  $L^*$  metrics strongly predict spectrophotometric total chlorophyll concentration in *Quercus robur* at both the leaf level ( $r = -0.81$ ,  $n = 168$ ) and the tree level ( $r = -0.95$ ,  $n = 7$ ). The species-specific calibration  $\text{Total Chl} = -0.501 \cdot b^* + 20.44$  ( $R^2 = 0.66$ ,  $\text{RMSE} = 1.61\ \mu\text{g mL}^{-1}$ ) provides a practical lookup for converting PhotoFolia  $b^*$  readings to estimated chlorophyll concentrations in the range  $5\text{--}20\ \mu\text{g mL}^{-1}$ . Cardinal direction has no systematic effect on PhotoFolia readings; a small proximal–distal leaf position effect is present but does not bias tree-level estimates. These results support the use of PhotoFolia as a field screening tool for *Q. robur* chlorophyll status in routine monitoring programmes, subject to the limitations noted regarding adaxial-only measurement and single-season calibration. The dataset and calibration parameters are provided to the PhotoFolia development team for inclusion in the species-specific validation database.

## References

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## **Appendix A. PhotoFolia field measurement protocol**

1. At the start of each session, take three test readings on a non-study leaf to confirm the application returns stable, expected colour values before data collection begins.
2. Use the same operator and the same device (Apple iPhone 10) throughout to minimise technique variability.
3. Work under diffuse or overcast light or in open shade. Postpone the session if direct sunlight prevents consistent imaging.
4. For each tree, photograph three leaves per cardinal direction (north, east, south, west).
5. For each leaf, image the proximal and distal positions on the adaxial (upper) surface, giving 24 readings per tree.
6. Hold the phone perpendicular to the leaf at the distance specified by the PhotoFolia protocol; avoid the midrib, major veins, damaged areas, and any specular reflection.
7. Discard and repeat any image the application flags as technically invalid.
8. Record  $L^*$ ,  $a^*$ ,  $b^*$ ,  $C^*$ ,  $h$ ,  $H$  and  $S$  for each photograph, with tree ID, crown position, leaf number, and adaxial position (proximal/distal).
9. Record the PhotoFolia application version at each session.

## **Appendix B. spectrophotometric chlorophyll extraction protocol**

1. Immediately after imaging, collect each photographed leaf into a labelled paper bag and transport to the laboratory in a cool box on the same day.
2. Punch a 10 mm diameter disc from each leaf and place it in a centrifuge tube with 3 mL of 80% v/v acetone.
3. Keep tubes in the dark at 4 °C for 24 hours to allow complete pigment dissolution. Prepare an 80% acetone solvent blank with each batch.
4. Centrifuge at 3000 rpm for 5 minutes and transfer the clear supernatant to a cuvette.

5. Measure absorbance at 645 nm and 663 nm on an Eppendorf BioSpectrophotometer (basic), against the solvent blank, within two hours of extraction.
6. Calculate Chl a, Chl b and total chlorophyll using the Lichtenthaler and Wellburn (1983) equations (Section 2.3).
7. Record raw absorbances, disc dimensions and extract volumes to allow full recalculation.

## **Appendix C. Raw data and code availability**

The raw paired dataset, the R analysis script, and the figures reproduced in this paper are available in the project repository. Links to be added:

Paired leaf-level data (PhotoFolia metrics + spectrophotometric chlorophyll, 168 rows):  
[add GitHub link]

Per-tree means: [add GitHub link]

Raw spectrophotometer absorbance readings (A645, A663): [add GitHub link]

R analysis script (correlations, calibration, figures): [add GitHub link]

Figures (PNG): [add GitHub link]